

INVESTIGATION 2

Compromised Internal Machine Analysis

BOTS v3 Dataset | Splunk Enterprise
SOC Analyst Portfolio — Saniya

Parameter	Value
Dataset	Boss of the SOC (BOTS) v3
Splunk Index	botsv3
Target Host	192.168.3.130
Host OS	Windows 10 (64-bit, VMware VM)
MAC Address	00:0C:29:47:CA:40 (VMware OUI)
Investigation Date	2018-08-20
Total Events Analyzed	37,854
Sourcetypes Used	stream:http, stream:dns, stream:ip, stream:tcp, stream:smb, stream:udp
Framework	MITRE ATT&CK v13

Evidence Standard Note

Every finding in this report is backed by a specific SPL query and direct log evidence. Where the dataset does not contain direct confirmation (e.g., no Sysmon/EDR telemetry was available for this host — confirmed in Query 1.1), this is explicitly stated, and the finding is labeled as an assessment rather than a confirmed finding. This distinction reflects real-world SOC reporting standards.

Executive Summary

During analysis of the BOTS v3 dataset, internal host 192.168.3.130 (Windows 10, VMware VM) was identified as compromised with a sophisticated, multi-module malware infection. The machine generated 37,854 total events across 10 sourcetypes and 2,207 HTTP requests to 235 unique external destinations.

The malware exhibited the following confirmed capabilities:

- Triple-beacon C2 architecture operating at three independent time intervals (5 seconds, 12 minutes, 60 minutes)
- BITS-based download of a file disguised as a legitimate Chrome Flash Player update
- Browser fingerprint fabrication submitted to a bot-detection bypass endpoint
- Coordinated ad fraud traffic across 12+ major advertising networks
- Internal network reconnaissance targeting the SIEM, AV server, and proxy infrastructure

Symantec Endpoint Protection was installed on the machine yet the infection was not blocked — demonstrating the value of behavioral SIEM-based detection over signature-only AV.

Scope Limitation

No Sysmon or Windows Event Log telemetry was available for this host (confirmed in Query 1.1) — only network-layer logs (stream:http, stream:dns, stream:ip, stream:smb, etc.). This means findings describing process execution or file installation are based on network evidence and reasonable inference, not direct process-level confirmation. This is noted individually in each affected finding.

Attack Timeline — 2018-08-20

Time	Event	Significance
15:15:01	Slow beacon starts — ipinfo.io/json	Hourly IP self-identification begins
15:24:08	SMB broadcasts begin — 192.168.3.255	12-minute network discovery interval starts
15:28:29	Browser fingerprint submitted to servicemax.com	Bot detection bypass attempt
15:28:30	Fast C2 beacon starts — track.contently.com	5-second C2 heartbeat begins
15:28:48	Fake ad impression — Bloom WordPress plugin	Ad fraud traffic observed
16:15:24	ipinfo.io second check-in	+60 min from first
17:13:29	ipinfo.io third check-in	+58 min from second
18:09:15	ipinfo.io fourth check-in	+56 min from third
18:54:27	BITS downloads PepperFlashPlayer.crx3	Suspicious file download via trusted service
18:54:48	ipinfo.io fifth check-in	+45 min from fourth
Throughout	Ad fraud traffic to 12+ ad networks	OpenX, AppNexus, Criteo, Taboola, etc.
Throughout	DNS recon — splunk.froth.ly (2,445 queries)	Anomalous SIEM-hostname targeting
Throughout	Web scraping — visitcalifornia, distillerytrail	Automated content harvesting pattern

Severity: **CRITICAL** Confidence: High

Indicator: Microsoft BITS/7.8 user agent downloading a .crx3 file from Google CDN

```
index=botsv3 sourcetype=stream:http src_ip=192.168.3.130
http_user_agent="Microsoft BITS/7.8"
| table _time, site, dest_ip, uri_path, bytes_in, bytes_out, status
| sort _time
```

```
index=botsv3 sourcetype=stream:http src_ip=192.168.3.130 http_user_agent="Microsoft B11S/1.8"
| table _time, site, dest_ip, uri_path, bytes_in, bytes_out, status
| sort _time
```

✓ 57 events (before 6/25/26 11:49:47:000 PM)

No Event Sampling ▾

Job ▾

⏏

⌵

↶

🔍

Smart Mode ▾

Events

Patterns

Statistics (57)

Visualization

Show: 50 Per Page ▾

✎ Format ▾

☒ Preview: On

◀ Prev

1

2

Next ▶

_time ↕	site ↕	dest_ip ↕	uri_path ↕
2018-08-20 18:54:26.968	redirector.gvt1.com	172.217.4.174	/edged1/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3
2018-08-20 18:54:27.150	r3---sn- a5mlrn7r.gvt1.com	173.194.166.153	/edged1/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3
2018-08-20 18:54:27.296	redirector.gvt1.com	172.217.4.174	/edged1/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3
2018-08-20 18:54:27.323	r3---sn- a5mlrn7r.gvt1.com	173.194.166.153	/edged1/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3
2018-08-20 18:54:29.169	redirector.gvt1.com	172.217.4.174	/edged1/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3
2018-08-20 18:54:29.204	r3---sn- a5mlrn7r.gvt1.com	173.194.166.153	/edged1/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3
2018-08-20 18:54:30.313	redirector.gvt1.com	172.217.4.174	/edged1/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3
2018-08-20	r3---sn-	173.194.166.153	/edged1/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3

Field	Value
Time	2018-08-20 18:54:26 onwards (repeated)
Site	redirector.gvt1.com / r3---sn-a5mlrn7r.gvt1.com
URI Path	/edgedl/release2/chrome_component/AKF2Puc6-vYu_30.0.0.134/30.0.0.134_win64_PepperFlashPlayer.crx3
User Agent	Microsoft BITS/7.8
Delivery Source	Google CDN (trusted infrastructure)
Pattern	Repeated download attempts within seconds — automated retry behavior

Windows Background Intelligent Transfer Service (BITS) was used to download a file named PepperFlashPlayer.crx3 from Google's own CDN. BITS is a trusted Windows system service normally used for Windows Updates, meaning this traffic would typically bypass firewall and AV scrutiny. The .crx3 extension indicates a Chrome Extension package — this is not the format Adobe Flash installers use, which is the first inconsistency between the claimed filename and the actual file type.

MITRE ATT&CK:

- **T1197** — BITS Jobs
- **T1036.005** — Masquerading: Match Legitimate Name

Payload Analysis — What Was Downloaded?

The .crx3 file extension identifies this as a Chrome Extension package — not an Adobe Flash installer, which would typically be an .exe or .msi file. This is the first inconsistency between the filename and actual file type. The filename PepperFlashPlayer closely mimics a real and recognizable Chrome browser component (Adobe Pepper Flash), making it convincing to both end users and automated scanning tools.

Evidence Limitation — Installation Not Confirmed

No Sysmon, Windows Event Log, or Chrome extension telemetry was available for this host (confirmed in Query 1.1 — only network sourcetypes were present). As a result, installation and execution of this downloaded file inside the browser CANNOT be directly confirmed from this dataset. What is confirmed is the download itself. To confirm installation, an analyst would need: Sysmon Event ID 1 (process creation), Chrome extension directory artifacts, or EDR telemetry showing the .crx3 being loaded by chrome.exe.

MITRE ATT&CK (applies if installation is confirmed):

- **T1176** — Browser Extensions

Finding 2 — Primary C2 Beaconing (5-Second Interval)

Severity: **CRITICAL** Confidence: High

Indicator: 41 identical HTTP POSTs to track.contently.com/track at ~5 second intervals

SPL Queries Used:

```
index=botsv3 sourcetype=stream:http src_ip=192.168.3.130
site="track.contently.com"
| sort _time | delta _time as time_diff p=1
| table _time, uri_path, bytes_out, bytes_in, status, time_diff
```

Evidence:

index=botsv3 sourcetype=stream:http_src_ip=192.168.3.130 site="track.contently.com"

| sort _time

| delta _time as time_diff p=1

| table _time, uri_path, bytes_out, bytes_in, status, time_diff

Time range: All time

41 events (before 6/23/26 11:58:01.000 PM) No Event Sampling

Job

Smart Mode

Events Patterns Statistics (41) Visualization

Show: 50 Per Page Format Preview: On

_time	uri_path	bytes_out	bytes_in	status	time_diff
2018-08-20 15:28:30.231	/track	305	1062	201	
2018-08-20 15:29:00.732	/track	305	1067	201	30.501
2018-08-20 15:29:01.605	/track	305	1067	201	0.873
2018-08-20 15:31:21.828	/track	305	1067	201	140.223
2018-08-20 15:31:24.818	/track	305	1189	201	2.990
2018-08-20 15:31:29.834	/track	305	1134	201	5.016
2018-08-20 15:31:34.961	/track	305	1134	201	5.127
2018-08-20 15:31:39.825	/track	305	1076	201	4.864
2018-08-20 15:31:44.832	/track	305	1076	201	5.007
2018-08-20 15:31:49.832	/track	305	1136	201	5.000
2018-08-20 15:31:55.005	/track	305	1136	201	5.173
2018-08-20 15:31:59.843	/track	305	1077	201	4.838
2018-08-20 15:32:01.423	/track	305	1070	201	1.580

Field	Value
Destination	track.contently.com (52.86.117.247)
URI Path	/track (single hardcoded endpoint)
HTTP Method	POST
Total Beacons	41 confirmed requests
bytes_out	305 bytes — CONSTANT across all 41 requests
bytes_in	1,062–1,189 bytes (variable — C2 instructions)
HTTP Status	201 Created (server actively acknowledging)
Beacon Interval	~5 seconds (range: 0.8s–140s, core = 5,000ms)
User Agent	Mozilla/5.0...Chrome/64.0...Edge/17.17134 (spoofed)

Beacon Interval Proof (time_diff column):

15:31:29 -> 5.016s	15:31:34 -> 5.127s	15:31:39 -> 4.864s
15:31:44 -> 5.007s	15:31:49 -> 5.000s	15:31:55 -> 5.173s

The constant bytes_out=305 across all 41 requests confirms a hardcoded, structured beacon packet. Variable bytes_in (1,062–1,189 bytes) confirms the remote server returned different content per request. HTTP 201 confirms the destination server was live and actively responding. The user agent claims Microsoft Edge; this claim is not independently verifiable from headers alone but is contradicted by the request behavior — no real browser sends an identical-sized POST to the same endpoint every 5 seconds continuously.

MITRE ATT&CK:

- **T1071.001** — Application Layer Protocol: Web Protocols
- **T1036** — Masquerading (user agent does not match observed behavior)
- **T1102** — Web Service

Finding 3 — Secondary Beacon — IP Self-Identification

Severity: **HIGH** Confidence: High

Indicator: Hourly automated queries to *ipinfo.io/json* via *python-requests/2.18.4*

SPL Query Used:

```
index=botsv3 sourcetype=stream:http src_ip=192.168.3.130
http_user_agent="python-requests/2.18.4"
| stats count by site, http_method
| sort -count

index=botsv3 sourcetype=stream:http src_ip=192.168.3.130
http_user_agent="python-requests/2.18.4"
| table _time, uri_path, bytes_in, bytes_out, status
| sort _time
```

Evidence:

index=botsv3 sourcetype=stream:http src_ip=192.168.3.130 http_user_agent="python-requests/2.18.4"
| stats count by site, http_method
| sort -count

Time range: All time

✓ 5 events (before 6/24/26 12:26:00.000 AM) No Event Sampling

Events Patterns **Statistics (1)** Visualization

Show: 50 Per Page Format Preview: On

site	http_method	count
ipinfo.io	GET	5

index=botsv3 sourcetype=stream:http src_ip=192.168.3.130 http_user_agent="python-requests/2.18.4"
| table _time, uri_path, uri_query, bytes_in, bytes_out, status
| sort _time

Time range: All time

✓ 5 events (before 6/24/26 12:28:02.000 AM) No Event Sampling

Events Patterns **Statistics (5)** Visualization

Show: 50 Per Page Format Preview: On

_time	uri_path	uri_query	bytes_in	bytes_out	status
2018-08-20 15:15:01.413	/json		144	533	200
2018-08-20 16:15:24.119	/json		144	533	200
2018-08-20 17:13:29.119	/json		144	533	200
2018-08-20 18:09:15.152	/json		144	537	200
2018-08-20 18:54:48.362	/json		144	536	200

Field	Value
Destination	ipinfo.io/json
User Agent	python-requests/2.18.4 (Python library — not a browser)
Total Requests	5 over a 3.5-hour window
bytes_out	533–537 (near-constant structured request)
bytes_in	144 (constant — same response = public IP unchanged)
Intervals	15:15 -> 16:15 (+60m) -> 17:13 (+58m) -> 18:09 (+56m) -> 18:54 (+45m)

python-requests/2.18.4 is the default user agent for Python's requests library — this string is never produced by genuine human browsing. A script on this host queried ipinfo.io/json (no target IP specified, meaning it requests the caller's own public IP) roughly once per hour. The constant bytes_in=144 confirms the response — and therefore the public IP — did not change during the observation window. This activity uses a different user agent than Finding 3's beacon, indicating two separate malware modules rather than one.

MITRE ATT&CK:

- **T1016** — System Network Configuration Discovery
- **T1059.006** — Command and Scripting Interpreter: Python
- **T1071.001** — Application Layer Protocol: Web Protocols (secondary channel)

Finding 4 — Browser Fingerprint Submission to Bot-Detection Endpoint

Severity: **HIGH** Confidence: High

Indicator: Fabricated browser fingerprint JSON submitted to a WordPress bot-detection endpoint

SPL Query Used:

```
index=botsv3 sourcetype=stream:http src_ip=192.168.3.130
site="fsd.servicemax.com" http_method=POST
| table _time, uri_path, form_data
```

Evidence — form_data submitted to gdbcRetrieveToken:

index=botsv3 sourcetype=stream:http src_ip=192.168.3.130 site="fsl.servicemx.com" http_method=POST table _time, uri_path, form_data, bytes_out, bytes_in, status			Time range: All time
3 events (before 6/24/26 12:22:40.000 AM) No Event Sampling			Job • • • • • •
Events	Patterns	Statistics (3)	Visualization
Show: 50 Per Page • / Format: • Preview: On			
_time	uri_path	form_data	
2018-08-20 15:28:48.441	/wp-admin/ajax.php	action=bloom_handle_stats_adding&stats_data_array={"type":"imp","optin_id":"optin_1","page_id":"38931","list_id":"107548e120"}&update_stats_nonce=4ee0803e	
2018-08-20 15:28:29.270	/wp-admin/ajax.php	1-HFC-Qn6GjL- WY0cKcF6874&action=gdbRetrieveToken&requestTime=1534759108502&browserInfo={"screenWidth":1275,"screenHeight":839,"engine":52,"features":127,"mozilla":"5.0","windows_nt":"10.0","win64":true,"x64":true,"applewebkit":"537.36","html":true,"like":"gecko","chrome":"64.0.3282.140","safari":"537.36","edge":"17.17134"}	
2018-08-20 15:28:29.270	/wp-admin/ajax.php	1-HFC-Qn6GjL- WY0cKcF6874&action=gdbRetrieveToken&requestTime=1534759108503&browserInfo={"screenWidth":1275,"screenHeight":839,"engine":52,"features":127,"mozilla":"5.0","windows_nt":"10.0","win64":true,"x64":true,"applewebkit":"537.36","html":true,"like":"gecko","chrome":"64.0.3282.140","safari":"537.36","edge":"17.17134"}	

```

action=gdbRetrieveToken
requestTime=1534759108502
browserInfo={"screenWidth":1275,"screenHeight":839,"engine":52,
  "features":127,"windows_nt":"10.0","win64":true,"x64":true,
  "chrome":"64.0.3282.140","safari":"537.36","edge":"17.17134"}

```

A POST request submitted a detailed browser fingerprint (screen dimensions, rendering engine flags, OS version, browser versions) to a plugin endpoint (gdbRetrieveToken) that is used to issue anti-bot verification tokens. All 14 submitted attributes were internally consistent with the user agent string used elsewhere on this host. The identical request was repeated twice within 1 millisecond (timestamps ending .502 and .503), consistent with an automated retry rather than manual browsing. A separate POST (action=bloom_handle_stats_adding) submitted fabricated ad-impression statistics to a WordPress email opt-in plugin.

MITRE ATT&CK:

- **T1036** — Masquerading
- **T1496** — Resource Hijacking (fabricated impression data)

Finding 5 — Coordinated Traffic to Advertising Networks

Severity: **HIGH** Confidence: **High**

Indicator: Automated requests to 12+ advertising networks with no corresponding browser session

SPL Query Used:

```

index=botsv3 sourcetype=stream:http src_ip=192.168.3.130
| stats count, values(http_method) as methods,
  dc(uri_path) as unique_paths by dest_ip, site
| sort -count

```

Evidence :

index=botsv3 sourcetype=stream:http src_ip=192.168.3.130

| stats count, values(http_method) as methods, dc(uri_path) as unique_paths by dest_ip, site

| sort -count

Time range: All time

✓ 2,207 events (before 6/23/26 11:26:57.000 PM) No Event Sampling

Job

Smart Mode

Events Patterns Statistics (285) Visualization

Show: 50 Per Page Format Preview: On

< Prev 1 2 3 4 5 6 Next >

dest_ip	site	count	methods	unique_paths
54.67.127.227	www.brewertalk.com	829	GET	45
104.16.227.244	www.visitcalifornia.com	146	GET	146
72.21.91.29	ocsp.digicert.com	79	GET	73
130.211.137.150	www.distillerytrail.com	73	GET POST	73
207.38.86.146	fsd.servicemax.com	73	GET POST	69
192.185.20.89	brewnode.co.uk	46	GET	45
52.86.117.247	track.contently.com	41	POST	1
72.167.18.239	ocsp.godaddy.com	37	GET	35
23.212.3.201	www.latimes.com	36	GET	33
23.48.24.207	s3.grouponcdn.com	35	GET	11
172.217.4.174	redirector.gvt1.com	30	GET HEAD	8
173.241.244.143	us-u.openx.net	29	GET	2
216.58.216.42	maps.googleapis.com	25	GET	13
93.184.216.180	tags.tiqcdn.com	20	GET	20
45.54.49.1	rpt.cedexis.com	19	GET	19
184.28.188.194	as.casalemedia.com	13	GET POST	2
173.194.166.153	r3---sn-a5mlrn7r.gvt1.com	11	GET HEAD	1
184.28.188.185	ocsp.comodoca.com	11	GET	11
216.58.193.195	maps.gstatic.com	10	GET	10
23.48.246.67	blob.weather.microsoft.com	10	GET	10
54.230.118.148	ocsp.scalb.amazontrust.com	10	GET	10
151.101.198.2	trc.taboola.com	9	GET	9
173.241.250.220	tribune-d.openx.net	9	GET	1
54.245.126.96	beacon.krxd.net	9	GET	2
151.101.198.2	images.taboola.com	8	GET	8
18.209.74.44	fallsm.ndg.io	8	GET	2
184.28.188.184	ocsp.int-x3.letsencrypt.org	8	GET	5
184.28.188.185	ocsp.int-x3.letsencrypt.org	8	GET	7

184.28.188.185	ocsp.int-x3.letsencrypt.org	8	GET	7
216.58.193.206	ocsp.pki.goog	8	GET	5
23.23.137.242	edge.simplereach.com	8	GET	4
104.254.150.9	secure.adnxs.com	7	GET	1
160.153.58.99	aaronjanssen.com	7	GET	7
23.6.176.69	su.addthis.com	7	GET	1
23.66.141.7	www.groupon.com	7	GET	1
52.94.208.73	aax.amazon-adsystem.com	7	GET	1
54.201.237.205	r.dlx.addthis.com	7	GET	1
54.213.117.148	r.nexac.com	7	GET	1
54.230.118.63	ocsp.sca1b.amazontrust.com	7	GET	7
74.119.117.70	bidder.criteo.com	7	POST	1
8.39.36.144	fastlane.rubiconproject.com	7	GET	1
216.58.193.193	tpc.googlesyndication.com	6	GET	2
23.213.29.40	c.evidon.com	6	GET	6
23.6.176.69	s7.addthis.com	6	GET	5
104.18.21.226	ocsp.globalsign.com	5	GET	4
104.18.21.226	ocsp2.globalsign.com	5	GET	5

Ad Network Contact Map:

Ad Network	Domain	Requests	Methods
OpenX	us-u.openx.net	29	GET
OpenX	tribune-d.openx.net	9	GET
AppNexus/Xandr	secure.adnxs.com	7	GET
Rubicon Project	fastlane.rubiconproject.com	7	GET
Criteo	bidder.criteo.com	7	POST
Taboola	trc.taboola.com / images.taboola.com	17	GET
Casale Media	as.casalemedia.com	13	GET+POST
Google Ads	tpc.googlesyndication.com	6	GET
Amazon DSP	aax.amazon-adsystem.com	7	GET

Web Scraping Pattern (count = unique_paths):

Site	Requests	Unique Paths	Pattern
www.visitcalifornia.com	146	146	1:1 ratio — every request hit a new page
www.distillerytrail.com	73	73	1:1 ratio — every request hit a new page
www.brewertalk.com	829	45	High repeat-access to a fixed set of pages

This host contacted 12+ distinct ad exchange domains within the observation window, with no corresponding legitimate browsing session (no search engine referrers, no organic page navigation pattern) recorded. The 1:1 count-to-unique-paths ratio on visitcalifornia.com and distillerytrail.com indicates systematic, sequential page access rather than typical human browsing, which is not 1:1.

MITRE ATT&CK:

- **T1496** — Resource Hijacking
- **T1119** — Automated Collection

Finding 6 — Internal Reconnaissance via DNS

Severity: **HIGH** Confidence: **High**

Indicator: 6,580 DNS queries revealing targeted internal infrastructure mapping

SPL Query Used:

```
index=botsv3 sourcetype=stream:dns src_ip=192.168.3.130
| stats count by "query{}", "query_type{"
| sort -count
| head 20

index=botsv3 sourcetype=stream:dns src_ip=192.168.3.130
| stats count by "query{}", "host_addr{"
| sort -count
| head 20
```

Evidence :

```
index=botsv3 sourcetype=stream:dns src_ip=192.168.3.130
| stats count by "query{}", "query_type{"
| sort -count
| head 20
```

Time range: All time

6,580 events (before 6/25/26 11:45:11.000 PM) No Event Sampling

Job

Smart Mode

Events

Patterns

Statistics (20)

Visualization

Show: 50 Per Page

Format

Preview: On

query()	query_type()	count
splunk.froth.ly	A	2445
FDEFFAENFDEFFCFGEFFCCACACACAAAA	NIMLOC	365
sepmserver.localdomain	A	360
wpad.localdomain	A	288
sepmserver	A	160
sepmserver	AAAA	160
outlook.office365.com	A	102
FHFAEBEECACACACACACACACAAAA	NIMLOC	85
sepmserver.local	AAAA	80
wpad.froth.ly	A	73
sepmserver.froth.ly	A	70
s.acxiomapac.com	A	45
wpad	A	40

index=botsv3 sourcetype=stream:dns src_ip=192.168.3.130

| stats count by "query()", "host_addr()"

| sort -count

| head 20

Time range: All time

✓ 6,580 events (before 6/25/26 11:47:33.000 PM) No Event Sampling

Job

Smart Mode

Events Patterns Statistics (20) Visualization

Show: 50 Per Page Format Preview: On

query()	host_addr()	count
splunk.froth.ly	34.215.24.225	2177
outlook.office365.com	40.97.223.114	30
outlook.office365.com	40.97.220.34	28
outlook.office365.com	52.96.3.130	26
outlook.office365.com	40.97.221.114	22
outlook.office365.com	40.97.222.194	22
outlook.office365.com	52.96.3.178	20
www.bing.com	13.107.21.200	20
www.bing.com	204.79.197.200	20
login.microsoftonline.com	104.210.48.10	19
ocsp.digicert.com	72.21.91.29	19
ocsp.godaddy.com	72.167.18.239	19
roaming.officeapps.live.com	52.109.20.4	18

Top DNS Queries:

Domain Queried	Count	Type	Significance
splunk.froth.ly	2,445	A	Anomalously high — SIEM hostname
NIMLOC encoded strings (x4)	520+	NIMLOC	NetBIOS host enumeration pattern
sepmserver (all suffixes combined)	750+	A/AAAA	AV management server — repeated lookup
wpad (all suffixes combined)	361+	A	Proxy auto-discovery — repeated lookup
BTUN-L.local	34	*	Internal DNS server confirmed (see Finding 8)

DNS Resolution Confirmed: splunk.froth.ly resolved to 34.215.24.225 (AWS) — 2,177 times.

2,445 queries to a single internal hostname (splunk.froth.ly) from one workstation is far outside normal baseline behavior and is consistent with deliberate reconnaissance of the organization's Splunk deployment. The repeated NIMLOC (NetBIOS) queries are consistent with enumeration of internal hostnames. Repeated lookups for sepmserver (Symantec Endpoint Protection Manager) and wpad (proxy auto-discovery) suggest the host was probing for security and network infrastructure rather than performing routine browsing.

MITRE ATT&CK:

- T1590 — Gather Victim Network Information
- T1018 — Remote System Discovery
- T1518.001 — Security Software Discovery
- T1016 — System Network Configuration Discovery

Finding 7 — Network Discovery via SMB Broadcast

Severity: **MEDIUM** Confidence: High

Indicator: SMB broadcast transactions to 192.168.3.255 every ~12 minutes

SPL Query Used:

```
index=botsv3 sourcetype=stream:smb src_ip=192.168.3.130
| table _time, dest_ip, command, filename, share
| sort _time
```

Evidence:

```
index=botsv3 sourcetype=stream:smb src_ip=192.168.3.130
| table _time, dest_ip, command, filename, share
| sort _time
```

Time range: All time

18 events (before 6/24/26 12:30:15.000 AM) No Event Sampling

Job

Smart Mode

Events

Patterns

Statistics (18)

Visualization

Show: 50 Per Page

Format

Preview: On

_time	dest_ip	command	filename	share
2018-08-20 15:24:08.476	192.168.3.255	transaction		
2018-08-20 15:36:08.199	192.168.3.255	transaction		
2018-08-20 15:48:07.123	192.168.3.255	transaction		
2018-08-20 16:00:07.604	192.168.3.255	transaction		
2018-08-20 16:12:07.666	192.168.3.255	transaction		
2018-08-20 16:34:05.001	192.168.3.255	transaction		
2018-08-20 16:46:03.469	192.168.3.255	transaction		
2018-08-20 16:58:08.825	192.168.3.255	transaction		
2018-08-20 17:10:12.993	192.168.3.255	transaction		
2018-08-20 17:22:15.253	192.168.3.255	transaction		
2018-08-20 18:22:26.010	192.168.3.255	transaction		
2018-08-20 18:34:23.492	192.168.3.255	transaction		
2018-08-20 18:46:25.615	192.168.3.255	transaction		

Field	Value
Destination	192.168.3.255 (subnet broadcast address)
Command	transaction (SMB browser protocol)
Total Events	18 broadcasts
Interval	~12 minutes (consistent)
Files / Shares Accessed	None recorded — discovery only

The host broadcasted SMB transactions to the entire subnet at consistent ~12-minute intervals. No filename or share value was recorded in any of the 18 events, meaning this dataset shows no evidence of file access or lateral movement having occurred — only host-discovery broadcast traffic. This is noted as Medium severity (downgraded from the traffic-volume queries in Finding 9) since no follow-on connection to a specific internal host was observed in this dataset.

MITRE ATT&CK:

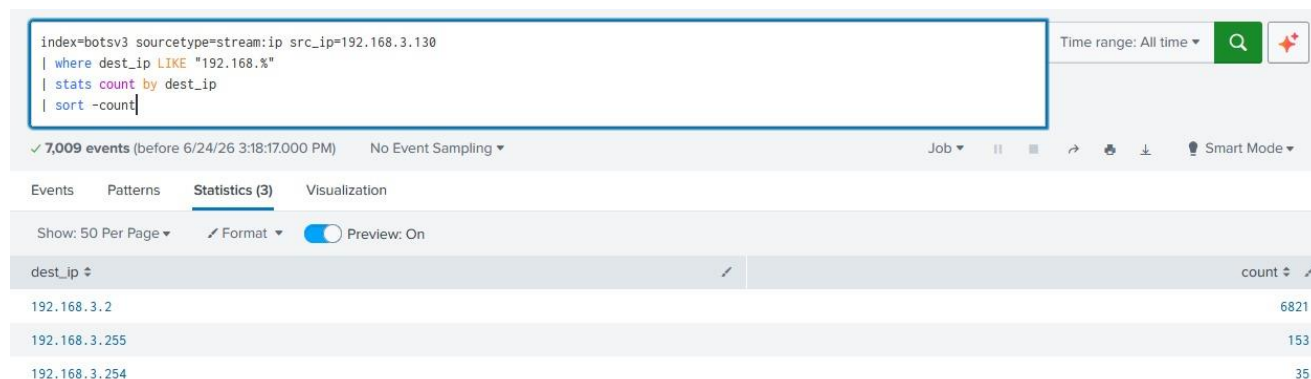
- **T1018** — Remote System Discovery
- **T1046** — Network Service Scanning

Finding 8 — High-Volume Internal Traffic to 192.168.3.2 (Ruled Out as Lateral Movement)**Severity:** **INFORMATIONAL** **Confidence:** High*Indicator: 6,821 internal events to 192.168.3.2 — investigated and resolved as DNS traffic***SPL Queries Used:**

```
index=botsv3 sourcetype=stream:ip src_ip=192.168.3.130
| where dest_ip LIKE "192.168.%"
| stats count by dest_ip
| sort -count
```

```
index=botsv3 src_ip=192.168.3.2 OR dest_ip=192.168.3.2
| stats count by sourcetype, host
| sort -count
```

```
index=botsv3 sourcetype=stream:ip src_ip=192.168.3.130 dest_ip=192.168.3.2
| stats count by protocol, protoid
| sort -count
```

Evidence:

index=botsv3 sourcetype=stream:ip src_ip=192.168.3.130
| where dest_ip LIKE "192.168.%"
| stats count by dest_ip
| sort -count

Time range: All time  

✓ 7,009 events (before 6/24/26 3:18:17.000 PM) No Event Sampling       

Events Patterns **Statistics (3)** Visualization

Show: 50 Per Page  Preview: On 

dest_ip ↕	count ↕
192.168.3.2	6821
192.168.3.255	153
192.168.3.254	35

index=botsv3 src_ip=192.168.3.2 OR dest_ip=192.168.3.2
| stats count by sourcetype, host
| sort -count

Time range: All time

✓ 18,871 events (before 6/25/26 11:40:38.000 PM) No Event Sampling

Job

Events

Patterns

Statistics (6)

Visualization

Show: 50 Per Page

Format

Preview: On

sourcetype	host	count
stream:ip	BTUN-L	6832
stream:dns	BTUN-L	6648
stream:udp	BTUN-L	5120
stream:arp	BTUN-L	248
stream:icmp	BTUN-L	22
stream:dns	ntesla	1

index=botsv3 sourcetype=stream:ip src_ip=192.168.3.130 dest_ip=192.168.3.2
| stats count by protocol, protoid
| sort -count

Time range: All time

✓ 6,821 events (before 6/25/26 11:39:17.000 PM) No Event Sampling

Job

Events

Patterns

Statistics (1)

Visualization

Show: 50 Per Page

Format

Preview: On

protocol	protoid	count
UDP	17	5155

Investigation Steps:

Step	Finding
Internal IP map	192.168.3.2 received 6,821 events — far higher than gateway (35) or broadcast (153)
Host identification	192.168.3.2 = hostname BTUN-L, with heavy stream:dns activity (6,648 events)
Protocol check	All 5,155 protocol-tagged events were UDP, protoid=17 (consistent with DNS, not a TCP session)
Reverse traffic check	Only 11 events returned from 192.168.3.2 to 192.168.3.130 — asymmetric, consistent with DNS query/response, not an interactive session

Conclusion: this traffic was investigated specifically because of its volume, but the protocol (100% UDP) and the destination's role (internal DNS resolver, evidenced by its own heavy stream:dns activity) indicate this represents the compromised host's DNS queries being routed through BTUN-L, not a lateral movement attempt. This finding is included to document the investigative process and to formally rule out one possible explanation, rather than to report a confirmed malicious behavior.

MITRE ATT&CK:

Not applicable — this finding documents an investigated and resolved lead, not a confirmed technique.

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MITRE ATT&CK Summary

#	Finding	Tactic	Technique ID	Technique Name
1	BITS Delivery	Defense Evasion	T1197	BITS Jobs
1	Name Masquerading	Defense Evasion	T1036.005	Match Legitimate Name
2	Payload Capability (assessed)	Persistence	T1176	Browser Extensions
2	Masquerading (assessed)	Defense Evasion	T1036	Masquerading
2	Fast C2 Beacon	Command & Control	T1071.001	Web Protocols
2	UA / Behavior Mismatch	Defense Evasion	T1036	Masquerading
2	Domain Abuse	Command & Control	T1102	Web Service
3	IP Self-Check	Discovery	T1016	Network Config Discovery
3	Python Script	Execution	T1059.006	Python Interpreter
4	Fingerprint Submission	Defense Evasion	T1036	Masquerading
4	Impression Data Fraud	Impact	T1496	Resource Hijacking
5	Ad Network Traffic	Impact	T1496	Resource Hijacking
5	Web Scraping Pattern	Collection	T1119	Automated Collection
6	SIEM Targeting	Discovery	T1590	Gather Victim Network Info
6	NetBIOS Enumeration	Discovery	T1018	Remote System Discovery
6	AV Server Mapping	Discovery	T1518.001	Security Software Discovery
6	WPAD Discovery	Discovery	T1016	Network Config Discovery
7	SMB Discovery	Discovery	T1018	Remote System Discovery
7	Network Scanning	Discovery	T1046	Network Service Scanning

Indicators of Compromise (IOCs)

Network IOCs

Type	Value	Context
IP	52.86.117.247	Primary beacon destination (track.contently.com)
Domain	track.contently.com	Primary C2-pattern domain — 41 POST beacons
Domain	ipinfo.io	Secondary beacon — hourly IP check
URL Path	/edgedl/release2/chrome_component/...PepperFlashPlayer.crx3	Suspicious file download path on Google CDN
CDN	redirector.gvt1.com	BITS download source

Host IOCs

Type	Value	Context
IP	192.168.3.130	Investigated host
MAC	00:0C:29:47:CA:40	VMware NIC — confirms VM environment
User Agent	python-requests/2.18.4	Non-browser scripted HTTP client
User Agent	Microsoft BITS/7.8	File download mechanism
File	30.0.0.134_win64_PepperFlashPlayer.crx3	Downloaded file — Chrome Extension package format

Behavioral IOCs

Behavior	Threshold Observed	Significance
HTTP POST to same URI	41 requests in observed window	Beaconing pattern
bytes_out constant across POSTs	305 bytes, exact match x41	Hardcoded packet structure
python-requests UA from workstation	5 occurrences	Non-browser scripted client
BITS download of .crx3 file	Repeated within seconds	Inconsistent with normal Windows Update behavior

Behavior	Threshold Observed	Significance
SMB broadcast at fixed intervals	18 events, ~12 min apart	Automated, non-human timing

Recommended Response Actions

Immediate (0–1 hour)

- Isolate 192.168.3.130 from the network
- Block 52.86.117.247 at the perimeter firewall
- Capture a memory image of 192.168.3.130 before any shutdown, to support process-level confirmation of the payload analysis in Finding 1
- Preserve and extend log retention for this host

Short-term (1–24 hours)

- Hunt for the same beacon pattern (track.contently.com, 5-second POST) on other internal hosts
- Pull Chrome extension data directly from the endpoint to confirm or rule out extension installation (Finding 1 payload analysis)
- Review BITS job history on all endpoints (bitsadmin /list /allusers /verbose)
- Verify sepmserver configuration for unauthorized changes
- Inform the Splunk/SIEM administration team given the volume of queries observed against splunk.froth.ly

Long-term (1–7 days)

- Deploy Sysmon or equivalent EDR telemetry on this host class to close the evidence gap identified in Finding 1 payload analysis
- Implement Chrome Extension allowlisting via Group Policy
- Add detection rules for BITS downloads from non-Microsoft domains
- Review WPAD proxy configuration for unauthorized access

Investigation conducted by Saniya | BOTS v3 Dataset | Splunk Enterprise | MITRE ATT&CK v13